Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 246 (2024) 473–481

Procedia Computer Science

www.elsevier.com/locate/procedia

28th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2024)

Usability study of gaze-based control methods in a game with time pressure

Katarzyna Kotyńska[a], Jacek Matulewski[b]*

[a]Student of Faculty of Philosophy and Social Science, Nicolaus Copernicus University, Toruń, Poland
[b]Department of Informatics, Faculty of Physics, Astronomy and Informatics, Nicolaus Copernicus University, Toruń, Poland

Abstract

A usability study of a gaze-controlled classic Breakout game with adaptive difficulty level, also taking into account the participation of a person using only gaze for communication and environment control, is presented in this article. Four variants of the gaze control method were tested, including two modifications: the way a paddle follows the gaze and the limitation of the gaze-aware area. The results show that in the first case, there is a discrepancy between what suits players better and what promotes better results. In the second case, limiting the gaze interaction area worsens both objectively and subjectively assessed usability. In addition, we checked whether experience in standard video games transfers to gaze-controlled games.

Peer-review under responsibility of the scientific committee of the 28th International Conference on Knowledge Based and Intelligent information and Engineering Systems

Keywords: gaze interaction; eye-tracking; arcade games; usability study

1. Introduction

1.1. Motivation

The long-term goal of software development and research conducted in our lab is to improve the usability of software package designed for disabled people who cannot operate computers using standard control methods [1]. The package includes text-based applications (instant messaging and therapeutic diary) [2-4] and "entertainment" applications (YouTube application, TV control application, board games including checkers, and set of classic

* Corresponding author. Tel.: +48 56 611 3310.
E-mail address: jacekmatulewski@umk.pl

1877-0509 © 2024 The Authors. Published by Elsevier B.V.

Peer-review under responsibility of the scientific committee of the 28th International Conference on Knowledge Based and Intelligent information and Engineering Systems
10.1016/j.procs.2024.09.427

arcade games including Breakout) [5, 6]. One of our goals is to alleviate the adverse effects of a patients' sudden breakdown in communication with their loved ones and loss of control over their life, and the resulting significant deterioration in well-being, as well as anxiety and depression. Hwang et al. [7] indicate that paralyzed individuals who utilize eye tracking as an assistive technology exhibit an improved quality of life compared to those who do not. Eye-tracking enhances social interaction capabilities and personal decision-making concerning oneself and one's care [8] and has the potential to be the most convenient alternative means of communication with a low entry threshold for individuals with various disabilities [9].

1.2. Gaze control methods

The common problem of gaze-controlled computer interfaces is an accidental interaction with it, which can easily occur when navigating using a gaze because of the dual role of the eye in such a situation - serving as both a sensor and a controller. This is known as the Midas touch problem. It can be solved by introducing a distinctive way of initiating the action. Techniques such as blinking [10], speech [11], and eyebrow movements [12] have been explored. However, these methods require control over muscles other than those controlling eyeball movements. Therefore, the most popular solution is to initiate an action after a more extended look (prolonged fixation) at the area related to this action. However, an excessive activation time strains the eyes and elongates the time required to perform intended tasks. It is also important to consider that placing, e.g., button labels or function descriptions on the buttons themselves may lead to unintended activations caused by the necessity to read them [13]. Moreover, this approach is less practical for games with time pressure, in which the time for selecting and approving actions is crucial for success, and fluidity of control is recommended.

An alternative method involves gaze gestures, specific gaze paths primarily composed of saccades. This technique is already employed, e.g., for text input [14, 15]. An attractive property of this approach is the ability to initiate gestures from any point on the screen and execute them at any scale [16, 17]. This may eliminate the need to calibrate the eye tracker. However, designing appropriate gestures is challenging due to the necessity of creating a balance between gestures' simplicity, associated with ease of learning, and complexity, preventing misidentification of natural eye movements, which requires extensive training [18]. According to the study of Istance [19], replicated by Hyrskykari et al. [20], it was found that individuals can quickly learn to perform reliable two- and three-legged gestures. It compares favorably to dwell-time-based control and eliminates the challenge of fixating on small objects. However, as a result of our pilot studies [21], game control methods based on two-legged gestures were rejected due to their poor results in usability tests, caused by their time consumption and eyestrain.

Another alternative type of gaze control utilizes a smooth pursuit, which occurs when the gaze follows a slowly moving object [22]. An example is the Dasher system [23] used for text input. In the case of Breakout, we could deal with this type of control if the paddle's position would be directly bound to the gaze position. Then, tracking the ball with the gaze would always move the paddle in the position to bounce the ball, which, however, causes the game to be devoid of the challenge that makes it attractive [24].

The gaze control method we employed differs from all the standard methods listed above. The paddle moves horizontally toward the gaze position only when it is on the paddle's left or right side. However, the movement speed is slow enough so the paddle does not follow the eye immediately. Thus, we avoid the above-described problem. We tested two variants, the one with a constant paddle speed and the one with a speed proportional to the distance of the paddle and gaze spot positions (cf. [25]). Moreover, in some experimental conditions paddle was gaze-sensitive only when the gaze was on the lower half of the screen.

During the pilot studies [21], we experimented with control methods allowing one to control the direction of the bounced ball depending on the hit part of the paddle. However, this turned out to be too difficult for novices. Therefore, in the described experiment, we used it only in the case of an expert and did not combine it with other control modifications.

1.3. Breakout

The project draws inspiration from the classic game Breakout, designed by Nolan Bushnell, Steve Wozniak, Steve Bristow, and Steve Jobs and published in 1976 by Atari. The game objective is to destroy several rows of bricks located at the upper side of the screen using a ball that can be bounced with a paddle located at the bottom of the screen. The paddle can only be moved horizontally. The need to control the movement of the paddle in only one dimension makes Breakout suitable for input modalities beyond traditional game controllers like a keyboard, gamepad, or computer mouse. Krepki et al. [26] adapted Breakout to be controlled using a brain-computer interface, and Dorr et al. [24] - to be controlled by gaze. In the latter case, the paddle position was bound directly to the user's gaze position, which significantly reduced the challenge, as described above.

2. Method

2.1. Software and details on gaze control method

The software was developed by Jacek Matulewski using Visual Studio 2019. It requires a Windows operating system and .NET Framework 4.7.2 or newer. WPF technology was used to prepare a graphical user interface (Fig. 1). It can work with eye trackers from several manufacturers, including: GazePoint, Tobii, SMI, Mirametrix etc.

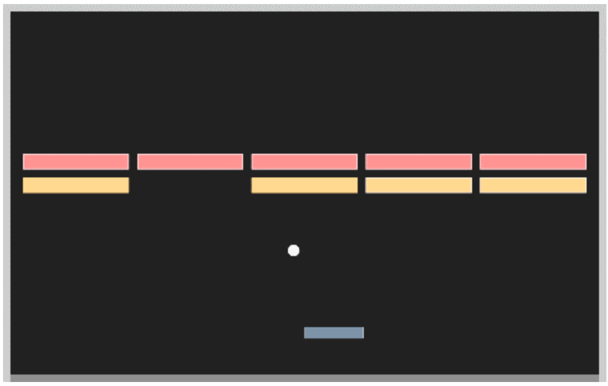

Fig. 1. Screenshot taken from the software used in research.

All implemented gaze control methods share the capability to pause the game by closing the eyes for two seconds and to terminate it if the eyes are closed for at least six seconds. During the game, the paddle is the only element controlled by gaze. It moves horizontally toward the x-coordinate of the gaze position, i.e., if it is to the left of the paddle's center, the paddle moves left, and vice versa. The research included two variants of this movement: with a constant speed of 2 px/ms and a speed proportional to the measured in pixels distance of gaze position from the paddle center (with a coefficient of 0.01). Both mentioned parameters were set quite arbitrarily by the authors, according to their subjective convenience during preliminary tests. However, it was verified that the speed of the paddle in both cases is lower than the maximum speed of smooth pursuit (about 30°/s), so that it can be tracked without catching-up saccades. In addition, other control method variation was also tested: with the gaze control area limited to the lower half of the screen (the paddle did not move when the player looked at the bricks) and the one without such limitation. In both cases, the player had to adopt a different ball-watching strategy. Therefore, four experimental conditions were applied in which we tested all four variants of control methods: method A - the paddle moves with a constant speed toward the gaze located anywhere on the screen, method B, which differs from A in the speed, which increases the farther the gaze is from the paddle, method C - the paddle moves at a constant speed as in method A, but only when the gaze is within the lower half of the screen, and method D - the paddle moves with speed scaling as in method B with limitation of gaze-aware area.

Additional method E was used only in the case study and not tested in the usability study. It follows the method's A way of moving the paddle but adds the ability to control the direction of the ball's bounce depending on what part of the paddle it hits. When bouncing from the very center of the paddle, the angle of reflection is equal to the angle of incidence. However, the closer to the side edges of the paddle the ball hits it, the greater the additional horizontal velocity component imparted to the batted ball. This is inconsistent with physics but increases the player's control over the game. However, at the expense of its complexity - efficient control of the speed after the bounce requires very precise control of the paddle's position at the moment of impact.

There is only one row of five bricks to destroy on the first game level. Each brick must be hit with the ball twice; after the first hit, the white frame around it disappears. Each successive level introduces one more row of bricks above. It was possible to place a maximum of five rows of bricks, but during the study, the game did not last long enough to reach this limit. Level numbers and elements related to achieved scores, which could distract participants, were removed during the study.

To maintain an optimal gaming experience and involvement of participants, an evaluation of the player's performance is conducted, and the game adjusts its difficulty based on the following algorithm. Every 10 seconds, the player's success coefficient is calculated, which is the quotient of successful ball bounces from the paddle to the number of all its passes down through the height at which the paddle is located. If its value is less than 0.25, the size of the tray is increased by 10%. However, if it is greater than 0.75, the palette is reduced by 10%. At the same time, the ball speed decreases or increases by 10%. This approach does not guarantee that a reduction and subsequent increase of paddle length or ball speed will restore the original value, but the differences are small enough. It should be noted that using this mechanism causes the comparison of points obtained by players or lives lost not to be reliable. Instead, the cumulative change in difficulty better reflects the control methods' usability.

Points were counted during the game: 1 point for each ball bounce with the paddle and 10 points for clearing the level and moving on to the next. Note, that there was no penalty for missing the ball (i.e. when it bounced off the bottom of the screen), but as mentioned above, these events affected the difficulty level.

2.2. Participants and location

25 students aged between 20 and 28 (M = 23.28, SD = 1.88) participated in the usability study. The gender distribution comprised 12 females, 11 males, and two individuals who preferred not to disclose their gender. All participants had normal vision or wore appropriate corrective glasses during the study. Those with visual impairments exhibited myopia ranging from -1 to -5 dioptres. Among the participants, 20 had no prior experience with eye-trackers, while five had participated in previous studies involving eye-tracking, but they don't use them on a daily basis. The study took place in a well-lit room using a laptop and the GazePoint GP3 HD eye-tracking device (sampling rate: 150 Hz). Participants were seated approximately 60 cm from the screen. Their head and body movements were not restricted. However, they were instructed to avoid excessive movement during the study to ensure consistent data collection.

The case study was conducted with a single participant, PD, who is affected by Amyotrophic Lateral Sclerosis (ALS) in the lateral stage, resulting in Quadriplegic Paralysis. The study was conducted in the participant's house, utilizing his personal laptop connected to a Tobii Pro eye-tracker (sampling rate: 60 Hz).

The usability study was conducted from January 16 to March 4, 2023. The case study research was concluded on May 25, 2022. The additional training was held from January 18 to January 30, 2023, with two males aged 22 and 23. All studies were conducted with Polish-speaking participants.

2.3. Procedure

We conducted three studies based on the above-described gaze-controlled Breakout game implementation, each with a unique focus. The main two are: a general usability study with healthy participants and a case study with an individual's participation in the lateral stage of ALS. While the core procedures were similar, each study had specific variations to address its distinct objectives. Additionally, a training study was conducted, which we treat as a pilot for subsequent studies, and the results of which will be presented only briefly.

Usability study. At the study initiation, participants were given an overview of the research procedure and informed of their option to interrupt the study at any point. To commence, participants completed an online questionnaire collecting personal information, followed by the eye-tracker calibration. Then, all control methods were presented to participants in a randomized order, accompanied by a concise textual description on paper. After reviewing the description, participants could pose questions. The core segment of the study involved 3 minutes of gameplay with an unlimited number of lives. Post-gameplay, participants completed the System Usability Scale (SUS) [27-29], consisting of 10 questions answered in the online form using a five-point Likert scale. Additionally, participants were invited to share feedback on the control method employed. After this, participants enjoyed a 3-minute break (with closed eyes) before evaluating the following control method. After testing all methods, participants were asked to complete a summary questionnaire and rank the control methods according to several categories (easiness, enjoyment, and frustration).

The usability study aimed to determine which gaze control methods are the most effective, user-friendly, and provide optimal gameplay experiences. Given the above considerations about gaze control methods and results of pilot studies [21], we suspect that the method utilizing constant speed will be easier to learn (it means that novice players will be less likely to lose lives and more likely to gain points, which leads to an increase in the difficulty level) but less rewarding. On the other hand, we suspect that limiting the gaze-control area leads to a more challenging game but also more satisfying. Therefore, we intended to validate the following hypotheses concerning the objective and subjective measures of usability:

H1a: In the case of control methods utilizing constant speed (i.e., methods A and C), the automatically adjusted difficulty level will increase more than in those with variable speed (B and D).

H1b: The usability of control methods utilizing variable speed (B and D) will be rated higher than those with constant speed (A and C).

H2a: In the case of control methods utilizing limited gaze-aware areas (C and D), the difficulty level will increase less than in those without such limitations (A and B).

H2b: Control methods utilizing limited gaze-aware areas (C and D) are more satisfying than those that do not utilize them (A and B).

Additionally, we wondered whether skills gained playing standard games controlled with typical game controllers profit in gaze-controlled games. We suspect that more time spent playing regular games, which is associated with greater gaming experience and practiced skills valuable in games, correlates with greater success in gaze-controlled games. We expressed this with the following hypothesis:

H3: Individuals who play computer games longer achieve higher scores in the Breakout average for all control methods.

Case study. PD engaged in gameplay for 6 minutes with each control method, but only the middle 3 minutes were considered in further analysis. It was because we observed PD needed about a minute to get used to the new gaze control method; we also wanted to equalize the length of the analyzed game recordings with those in usability studies for possible comparison. The surveying procedure had to be adapted to accommodate his needs. As PD is non-verbal, an augmentative and alternative communication table was used to facilitate responses to questions. A colorful communication table, inclusive of a Likert scale and options for yes/no responses, was designed. Each question was read aloud, and based on PD's gaze position, along with the assistance of a caregiver, their assessment of gaze control methods was recorded.

Training study. Additionally, a pilot longitudinal study was conducted with two participants training all control methods to check whether preferences for gaze control may change over time. Participants were tasked with four ten-minute training sessions every 2-3 days, each utilizing a different control method. At the beginning and after completing the training, participants were subjected to the same procedure as in the usability study to compare pre- and post-training results.

3. Results

3.1. Usability study

Subjective measures. In the usability study, method B emerged as the most useful control method in participants' opinion, obtaining the highest average score in the SUS survey (M = 80.9 points, SD = 17.37) (cf. Fig. 1a). Notably, 13 out of 25 participants identified method B as the most enjoyable in the final ranking. Method A received a mean SUS questionnaire score of 76.8 points (SD = 17.77), method D scored an average of 71.2 points (SD = 18.70), and method C obtained 64.4 points (SD = 20.28). However, the differences between methods A and B and methods C and D are statistically insignificant (here and below, the non-parametric Mann-Whitney test was used due to the lack of normality of the distribution), contradicting hypothesis H1b. Thus, the way of changing the speed does not affect the perceived usefulness of control methods. The comparison of methods A and B in terms of their difficulty declared by the participants was also not conclusive since both methods were equally regarded as the easiest by participants, with ten votes each.

For a change, the comparison of methods with and without the limited gaze-aware area shows the differences. A paired comparison of methods A and C and methods B and D shows that SUS score mean values differences are statistically significant ($p < 0.05$). However, contrary to our assumptions, the methods with a limited interaction area lowered the usability score, not increased it. Therefore, hypothesis H2b is also rejected.

It must be noted that the SUS survey ratings for individual methods were very diverse. For instance, despite receiving a score of around 32.5 points from one user, method D also had a group of users who rated it 100 points. Thus, it is difficult to talk about choosing the one best method but rather about selecting the methods that suit the preferences of individual players.

Objective measures. When utilizing method A, players achieved the highest average final score in the game (M = 94.33 points, SD = 24.76), compared to the lowest one for method C (M = 87.24, SD = 18.5) (cf. Fig. 2a), which was also indicated as the most frustrating in the final ranking. Notably, both the highest and lowest personal scores were obtained for method A (191 and 53 points). Method A also stood out with the lowest number of lives lost (mean value = 25.44, SD = 10.2) (cf. Fig. 2c) compared to the highest mean value for method D (M = 32.6, SD = 9.4). However, as we pointed out above, comparing these results may be unreliable due to the adaptive adjustment of the difficulty level (by changing the paddle's size and the ball's speed). Thus, it seems more appropriate to compare the cumulative change in difficulty (Fig. 2d). Note that a similar mean number of points obtained for various gaze control methods (Fig. 2b) suggests that the difficulty adaptation algorithm is working reasonably. Comparing the final change in the difficulty level, it increased the most in the case of method A (M = 6.84), which confirms its largest usability. It is much higher than for both methods B (M = 4.8) and C (M = 4.6), and the differences are statistically significant in both cases ($p < 0.05$ for A and B and for A and C). It means that both the speed modification and the limitation of the gaze-aware area, reduce the objectively measured usability of gaze control methods. It is even worse if method A is compared to method D, which utilizes both modifications ($p < 0.001$). It confirms hypothesis H1a but contradicts hypothesis H2a. Thus, method A produced the best results (the largest change in the difficulty level, but also the highest number of points obtained and the least lives lost), however, this is the method B, which was the best rated in a subjective assessment.

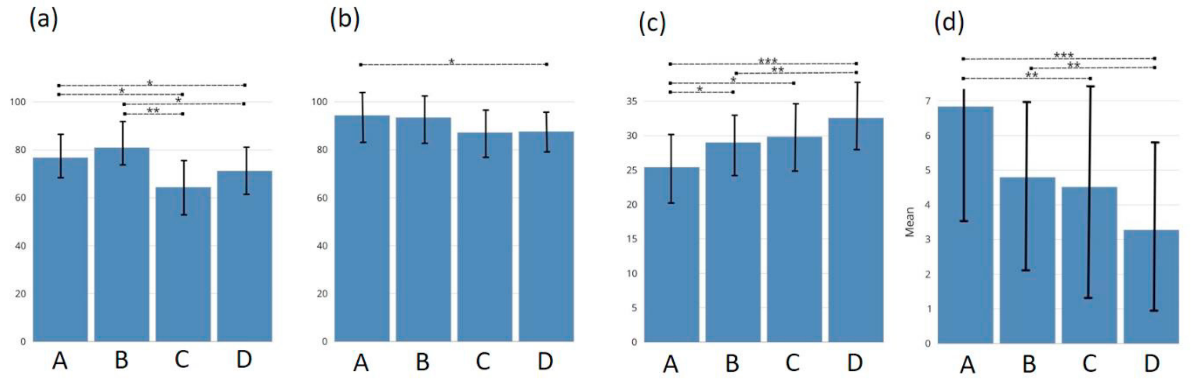

Fig. 2. (a) The mean results of the SUS questionnaire for all methods (marked with capital letters under the bars), (b) the mean final score, (c) the mean lost lives, and (d) the mean accumulated difficulty level change.

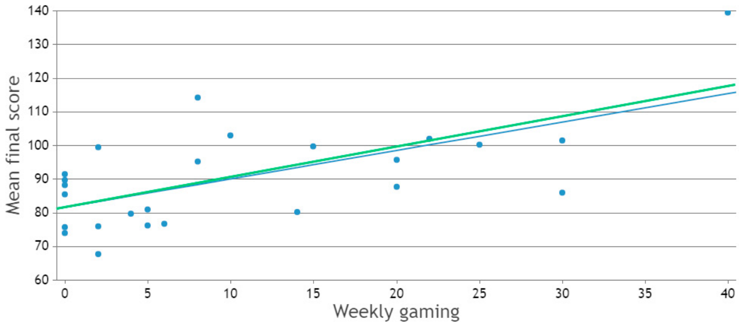

Fig. 3. Scatter plot showing a trendline between the mean final score and weekly hours spent playing video games. The blue line is the trend for every participant, and the green one excludes people not playing games.

The analysis of the points gained in the examined game and the declared by participants number of hours spent playing games per week uncovered a statistically significant one-tailed Spearman's rank correlation ($\rho = 0.527$, $p < 0.001$; cf. Fig. 3), which supports hypothesis H3. The correlation was still significant when people who did not play video games at all were excluded ($\rho = 0.635$, $p < 0.01$).

3.2. Case study

To reduce the PD's effort, this study tested only three methods: B, C, and E. According to the SUS questionnaire, method B received the highest rating with 77.5 points, closely followed by method C with 75 points. For comparison, the corresponding average ratings in the usability study were 80.9 and 64.4, respectively. Method E scored significantly lower, with only 37.5 points. Interestingly, despite its low usability rating, method E achieved the highest final score of 87. The number of lost lives was the same for methods B and C, totaling 26, while method E resulted in 31 lives lost. Finally, the cumulative change of difficulty level revealed a substantial gap between method E, for which it is equal to 1, and methods B and C, which scored 5 and 7, respectively. For the latter two methods, the increase in difficulty was greater than the average increase in the case of usability study participants, probably due to the PD's greater experience in gaze-controlled interfaces.

3.3. Training

As mentioned, the number of training participants was too small for the results to be worth presenting in detail. Nevertheless, we would like to point out some interesting facts. The average ratings in the SUS survey for methods A and D did not change much after training (46.25 vs. 46.25 for method A, 71.25 vs. 70.00 for D), but the rating for method B decreased by almost half (78.75 vs. 46.25), and the rating for method C almost doubled (42.50 vs. 81.25). Interestingly, the final level of difficulty increased significantly in the case of both of these methods (5 vs. 9.5 and 6 vs. 11.5) and much less in the case of method A (11 vs. 12.5). However, in the case of method D - it even decreased (8.5 vs. 7.5). This shows that gaze control method preferences do not have to be permanent. They may change as the player becomes more experienced with a particular game or gaze control in general.

4. Conclusions

We explored several variants of gaze control in the Breakout game, focusing on its two modifications. The first one involved limiting the gaze-aware area, and the second concerned how the paddle moved depending on the distance from the gaze position on the screen. Concerning the first one, a subjective usability assessment showed that the study participants liked the control method more, in which the paddle moves faster the farther the gaze position is from it. However, objective measures have not confirmed its greater usability. In particular, the degree of change in the automatically adjusted difficulty increased more for the method with constant paddle speed. The second modification is more compatible. Limiting the gaze-aware area neither appeals to players nor helps them achieve better results in the game. Moreover, the correlation between superior game performance and higher gaming frequency underscores the significance of prior gaming experience in mastering gaze-controlled gameplay. In addition, participants' evolving preferences during extended gameplay sessions suggest the potential influence of familiarity and experience on users' perceptions of different control methods. This highlights the dynamic nature of user preferences and the need for adaptive systems that cater to individual learning curves.

The conclusions regarding paddle control in Breakout game can be generalized to one-dimensional control of other game elements (e.g. determining the direction of a vehicle, steering a tank turret, controlling characters moving on a tracks, as in games such as Subway Surfers and many others), and also extended to the control of sliders (e.g. in many kinds of players for volume control or indicating the movie fragment).

Acknowledgements

The paper has been developed thanks to the help of the Centre for Modern Interdisciplinary Technologies, Nicolaus Copernicus University in Torun, Poland.

References

[1] Kalyanpur, M., and Kirmani, M. H. (2005). "Diversity and technology: Classroom implications of the digital divide." *Journal of Special Education Technology* **20(4)**: 9-18.

[2] Matulewski, J., and Patera, M. (2020) "Comparison of three dwell-time-based gaze text entry methods: Extended Abstract." *ACM Symposium on Eye Tracking Research and Applications* (ETRA '20 Adjunct). Association for Computing Machinery, New York, NY, USA, **Article 15**, 1–5.

[3] Matulewski, J., and Patera, M. (2022) "Usability of the super-vowel for gaze-based text entry." *ETRA '22: 2022 Symposium on Eye Tracking Research and Applications*. Association for Computing Machinery, New York, NY, USA, **Article 69**, 1–5.

[4] Szwarc, J., Rasmus, A., Koziara, J., and Matulewski, J. (2023) "What is supposed to help, actually hinders: about the impact of the GUI dynamics on the efficiency of gaze text entry." *Procedia Computer Science* **225C**: 2291–2300.

[5] Matulewski, J., Bałaj, B., Marek, E., Piasecki, Ł., Gruszczyński, D., Kuchta, M., and Duch, W. (2018) "Moveye: gaze control of video playback" *ACM Proceedings of the Workshop on Communication by Gaze Interaction* (ETRA, COGAIN'18).

[6] Grech, J., and Matulewski, J. (2023) "Evaluation of several gaze control methods for a board game with no time pressure." *Procedia Computer Science* **225C**: 2456–2465.

[7] Hwang, C. S., Weng, H. H., Wang, L. F., Tsai, C. H., and Chang, H. T. (2014). "An eye-tracking assistive device improves the quality of life for ALS patients and reduces the caregivers' burden." *Journal of Motor Behavior* **46(4)**: 233-238.

[8] Kivimäki, M., Vahtera, J., Elovainio, M., Helenius, H., Singh-Manoux, A., and Pentti, J. (2005). "Optimism and pessimism as predictors of change in health after death or onset of severe illness in family." *Health psychology* **24(4)**: 413.

[9] Donegan, M., Morris, J. D., Corno, F., Signorile, I., Chió, A., Pasian, V., and Holmqvist, E. (2009) "Understanding users and their needs." *Universal Access in the Information Society* **8(4)**: 259-275.

[10] Missimer, E., and Betke, M. (2010). "Blink and wink detection for mouse pointer control." in *Proceedings of the 3rd International Conference on PErvasive Technologies Related to Assistive Environments* (pp. 1-8).

[11] Kaur, M., Tremaine, M., Huang, N., Wilder, J., Gacovski, Z., Flippo, F., and Mantravadi, C. S. (2003). "Where is" it"? Event synchronization in gaze-speech input systems." in *Proceedings of the 5th International Conference on Multimodal Interfaces* (pp. 151-158).

[12] Grauman, K., Betke, M., Lombardi, J., Gips, J., and Bradski, G. R. (2003). "Communication via eye blinks and eyebrow raises: Video-based human-computer interfaces." *Universal Access in the Information Society* **2(4)**: 359-373.

[13] Penkar, A. M., Lutteroth, C., and Weber, G. (2012). "Designing for the eye: design parameters for dwell in gaze interaction." in *Proceedings of the 24th Australian Computer-Human Interaction Conference* (pp. 479-488).

[14] Perlin, K. (1998) "Quikwriting: continuous stylus-based text entry." in *Proceedings of the 11th annual ACM symposium on User interface software and technology* (pp. 215-216).

[15] Wobbrock, J. O., Rubinstein, J., Sawyer, M. W., and Duchowski, A. T. (2008). "Longitudinal evaluation of discrete consecutive gaze gestures for text entry." in *Proceedings of the 2008 Symposium on Eye Tracking Research & Applications* (pp. 11-18).

[16] Drewes, H., and Schmidt, A. (2007). "Interacting with the computer using gaze gestures." in *IFIP Conference on Human-Computer Interaction* (pp. 475-488). Springer, Berlin, Heidelberg.

[17] Drojecki, P., Sobczuk, W., and Matulewski, J. (2022) "Gaze Buttons Revisited. Looking for an Effective Way of "Clicking" without the Mouse". *Procedia Computer Science* **207**: 1539–1551.

[18] Köpsel, A., Majaranta, P., Isokoski, P., and Huckauf, A. (2016). “Effects of auditory, haptic and visual feedback on performing gestures by gaze or by hand.” *Behaviour & Information Technology* **35(12)**: 1044-1062.

[19] Istance, H., Hyrskykari, A., Immonen, L., Mansikkamaa, S., and Vickers, S. (2010). “Designing gaze gestures for gaming: an investigation of performance.” in *Proceedings of the 2010 Symposium on Eye-Tracking Research & Applications* (pp. 323-330).

[20] Hyrskykari, A., Istance, H., & Vickers, S. (2012). “Gaze gestures or dwell-based interaction?” in *Proceedings of the Symposium on Eye Tracking Research and Applications* (pp. 229-232).

[21] Kotyńska, K. (2021), “Design of gaze interaction in games with time pressure. Pilot studies.”, diploma thesis, https://apd.umk.pl/diplomas/134861/

[22] Vidal, M., Bulling, A., and Gellersen, H. (2013). “Pursuits: spontaneous interaction with displays based on smooth pursuit eye movement and moving targets.” in *Proceedings of the 2013 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (pp. 439-448).

[23] Ward, D. J., Blackwell, A. F., and MacKay, D. J. (2000). “Dasher - a data entry interface using continuous gestures and language models.” in *Proceedings of the 13th Annual ACM Symposium on User Interface Software and Technology* (pp. 129-137).

[24] Dorr, M., Böhme, M., Martinetz, T., and Barth, E. (2007). “Gaze beats mouse: A case study.” in *Proceedings of COGAIN In The Proceedings of 3rd Conference on Communication by Gaze Interaction* (COGAIN 2007).

[25] Stellmach, S., & Dachselt, R. (2012). “Designing gaze-based user interfaces for steering in virtual environments.” in *Proceedings of the Symposium on Eye Tracking Research and Applications* (pp. 131-138).

[26] Krepki, R., Blankertz, B., Curio, G., and Müller, K. R. (2007). “The Berlin Brain-Computer Interface (BBCI) - towards a new communication channel for online control in gaming applications.” *Multimedia Tools and Applications* **33(1)**: 73-90.

[27] Brooke, J. (2013). “SUS: a retrospective.” *Journal of Usability Studies* **8(2)**: 29-40.

[28] Bangor, A., Kortum, P. T., and Miller, J. T. (2008). “An empirical evaluation of the system usability scale.” *International Journal of Human–Computer Interaction* **24**: 574–594.

[29] Sauro, J., and Lewis, J. R. (2016). “Quantifying the user experience: Practical statistics for user research.” Morgan Kaufmann.